

Income and Walkability in the NC Triangle

Joshua Mark

Department of Geography, The University of North Carolina at Chapel Hill

GEOG 515: Applied Spatial Data Science

Dr. Paul L. Delamater

May 4, 2026

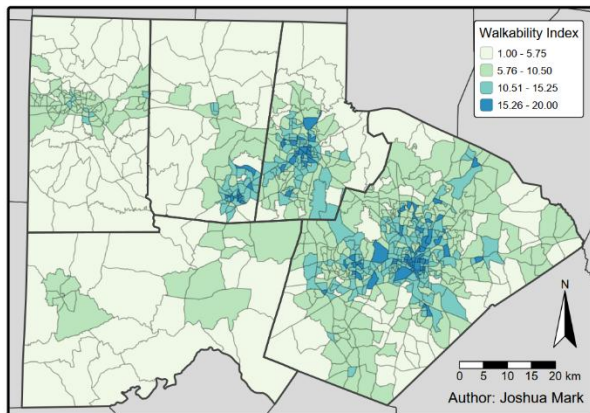
Introduction

The NC Triangle region is often credited for its walkability, but what factors influence that walkability, and is there a pattern to its distribution? This project aims to understand the relationship between a community's level of income and their access to walkable areas. The project asks: "How is the percentage of low-to-moderate income households related to walkability within the NC Triangle, after adjusting for population density and spatial autocorrelation?" The hypothesis is that more affluent communities will have better access to walkable areas, which would result in a strong negative correlation between the variables. The purpose of the project is to investigate potential inequality in access to walkable areas.

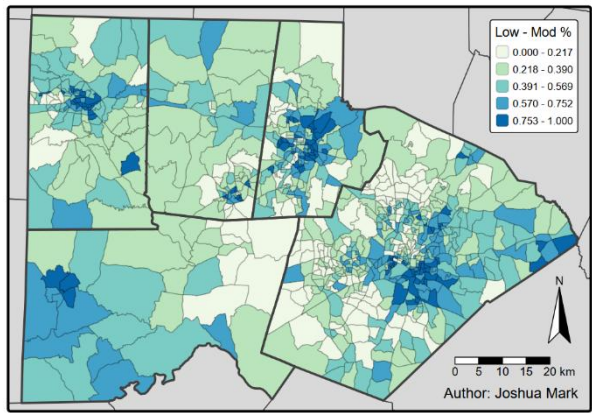
Data and Methods

This project utilizes two main datasets: the EPA's National Walkability Index and the HUD's Low to Moderate Income (LMI) Populations dataset. Both are on a block group scale. The walkability data, provided by the EPA, was originally prepared in 2017 and was most recently updated in 2021, reflecting 2019 census block groups. Areas are scored on a scale of 1 to 20 and are rated based on a variety of factors including intersection density, proximity to transit stops, and diversity of land uses. The next dataset is procured from the Department of HUD. It covers ACS data between 2016 and 2020 and is recorded in 2020 census block groups. Data on income, such as median household income, is typically not collected at the block group scale. This dataset instead sorts people into low-, medium-, and high-income categories, and from there, it can measure the percent of households within a block that meet these criteria. Other data utilized includes county shapefiles acquired through R's Tigris package.

Walkability in the NC Triangle



Low - Moderate Income Households in the NC Triangle



Some basic cleaning steps done to both datasets include filtering to the triangle area, selecting relevant columns, and reprojecting to EPSG 32119. Under the walkability dataset, population density was calculated by dividing population data by an area measurement. The most notable step taken was combining the datasets for comparison. This was complicated because of the incongruity of 2019 and 2020 block groups. To resolve this, the spatial intersection between them was calculated and the more divided 2020 block groups, hosting the LMI data, were aggregated to fit the walkability dataset. This was done by finding the area of each intersection, calculating a weighted metric of LMI based on that area, and grouping the areas by 2019 block groups. a final area-weighted average for LMI was summarized into each block group. These values were used in the analysis.

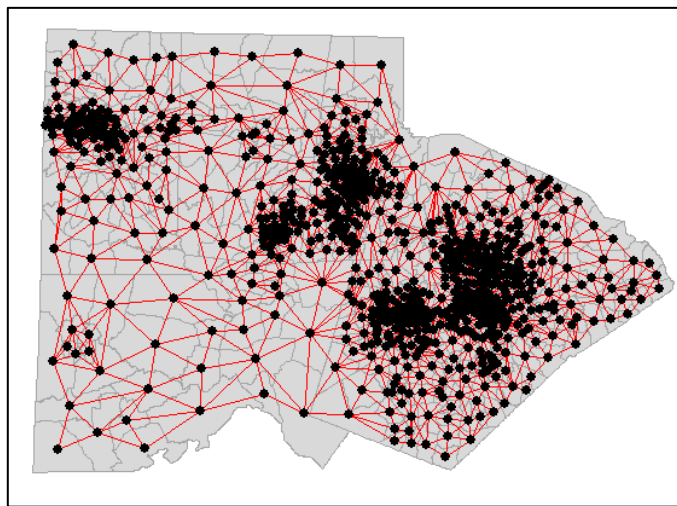
The analysis begins with a preliminary exploration into the variables. First, a neighborhood definition is created for the area and Moran's I tests are run on every variable. Neighborhoods are defined using queen's 1st order. Running Moran's I on each variable is not a necessary step for the final analysis, but it is a good step to take for exploring the statistics. From here, the model is built step by step, starting at walkability ~ pop_density and finally, walkability ~ pop density + low_mod_pct. Next, a Moran's I test is performed on the full model's residuals.

If spatial autocorrelation is significant, the regression model must be scrapped in favor of a spatial model.

Assuming spatial autocorrelation is found, the next step is to run a Lagrange Multiplier (LM) test to determine whether a lag or error model is best (and potentially a SARMA model). After determining necessary models and running them, the final step is to check once more for evidence of spatial autocorrelation, and if it is insignificant, the results can be interpreted.

Results

Preliminary outputs:



```
> tri_q1nb
Neighbour list object:
Number of regions: 831
Number of nonzero links: 5178
Percentage nonzero weights: 0.7498251
Average number of links: 6.231047
```

Top: Output of neighborhood object summary.

Left: map of Queen's 1st order neighborhood connections in the triangle.

```
Moran I test under randomisation

data: tri_walk_LMI$WALK_INDEX
weights: tri_q1nb_rs

Moran I statistic standard deviate = 32.148, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic      Expectation      Variance
    0.6427941009      -0.0012048193      0.0004012942
```

Result of Moran's I test for the walkability variable.

```
Moran I test under randomisation

data: tri_walk_LMI$low_mod_pct
weights: tri_q1nb_rs

Moran I statistic standard deviate = 22.798, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic      Expectation      Variance
    0.4554661678      -0.0012048193      0.0004012346
```

Result of Moran's I test for the low-moderate income percent variable.

```
Moran I test under randomisation

data: tri_walk_LMI$POP_DENS_KM2
weights: tri_q1nb_rs

Moran I statistic standard deviate = 24.406, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic      Expectation      Variance
    0.4853494343      -0.0012048193      0.0003974295
```

Result of Moran's I test for the population density variable.

Initial regression model:

```
Call:
lm(formula = WALK_INDEX ~ POP_DENS_KM2 + low_mod_pct, data = tri_walk_LMI)

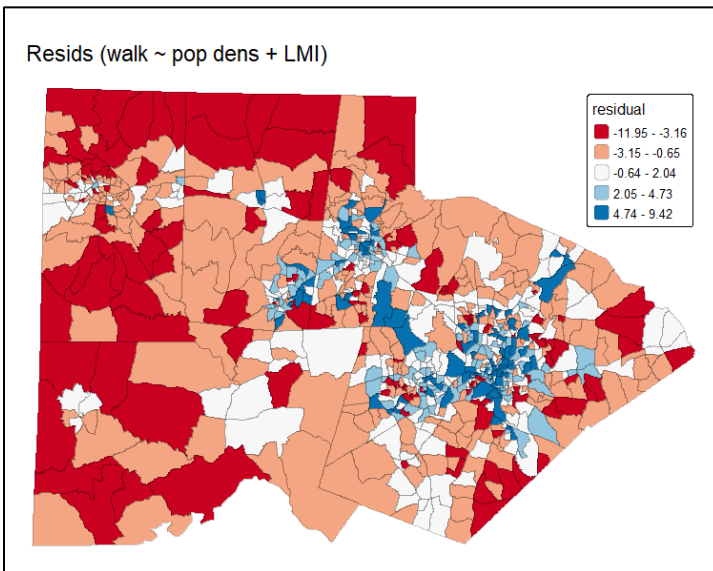
Residuals:
    Min       1Q   Median       3Q      Max
-11.9465  -2.4399  -0.4483   2.0301   9.4176

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.4404076  0.2444987  26.341  <2e-16 ***
POP_DENS_KM2  0.0022978  0.0001268  18.118  <2e-16 ***
low_mod_pct   2.1077846  0.5022714   4.197   3e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.275 on 828 degrees of freedom
Multiple R-squared:  0.3437,    Adjusted R-squared:  0.3421
F-statistic: 216.8 on 2 and 828 DF,  p-value: < 2.2e-16
```

Significance ($p < .05$) found in each variable.

Note: results here are meaningless if residuals show evidence of autocorrelation.



Map of model residuals.
Visible evidence of clustering / spatial autocorrelation.

```
Moran I test under randomisation

data: tri_walk_LMI$walk_dens_LMI_resid
weights: tri_q1nb_rs

Moran I statistic standard deviate = 18.882, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:
Moran I statistic      Expectation      Variance
    0.3768949294      -0.0012048193      0.0004009744
```

Moran's I on residuals. Proven global spatial autocorrelation. Spatial model required.

Spatial Models:

```
Rao's score (a.k.a Lagrange multiplier) diagnostics for spatial dependence
data:
model: lm(formula = WALK_INDEX ~ POP_DENS_KM2 + low_mod_pct, data = tri_walk_LMI)
test weights: tri_qlnb_rs

      statistic parameter p.value
RSerr    351.7088          1 < 2e-16 ***
RSlag    525.2587          1 < 2e-16 ***
adjRSerr    3.0138          1 0.08256 .
adjRSlag   176.5638          1 < 2e-16 ***
SARMA     528.2726          2 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Top: Lagrange Multiplier (LM) test. Lag and error are equally significant. Adjusted error model loses significance. SARMA is significant. Models to run are lag and SARMA.

```
Call:lagsarlm(formula = WALK_INDEX ~ POP_DENS_KM2 + low_mod_pct, data = tri_walk_LMI,
  listw = tri_qlnb_rs, zero.policy = TRUE)

Residuals:
    Min       1Q   Median       3Q      Max
-6.74213 -1.56163 -0.19717  1.40056  7.85947

Type: lag
Coefficients: (asymptotic standard errors)
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.22390142  0.26966131  4.5387 5.661e-06
POP_DENS_KM2  0.00081643  0.00009922  8.2285 2.220e-16
low_mod_pct   1.51243401  0.37261811  4.0589 4.930e-05

Rho: 0.71881, LR test value: 424.94, p-value: < 2.22e-16
Asymptotic standard error: 0.028269
z-value: 25.427, p-value: < 2.22e-16
Wald statistic: 646.55, p-value: < 2.22e-16

Log likelihood: -1950.92 for lag model
ML residual variance (sigma squared): 5.7219, (sigma: 2.392)
Number of observations: 831
Number of parameters estimated: 5
AIC: 3911.8, (AIC for lm: 4334.8)
LM test for residual autocorrelation
test value: 22.762, p-value: 1.8331e-06
```

Spatial Lag Model.

Models shows significance ($p < .05$) in both population density and low_mod_pct variables.

LM test for residual autocorrelation comes back with significant autocorrelation.

```
Call:sacsarlm(formula = WALK_INDEX ~ POP_DENS_KM2 + low_mod_pct, data = tri_walk_LMI,
  listw = tri_qlnb_rs, type = "sac", zero.policy = TRUE)

Residuals:
    Min       1Q   Median       3Q      Max
-6.52428 -1.44878 -0.20234  1.33038  7.35593

Type: sac
Coefficients: (asymptotic standard errors)
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.6395e-01  2.1853e-01  2.1231 0.0337492
POP_DENS_KM2  6.7060e-04  9.1108e-05  7.3605 1.832e-13
low_mod_pct   1.0676e+00  2.9755e-01  3.5881 0.0003331

Rho: 0.83478
Asymptotic standard error: 0.028016
z-value: 29.797, p-value: < 2.22e-16
Lambda: -0.43999
Asymptotic standard error: 0.090388
z-value: -4.8678, p-value: 1.1287e-06

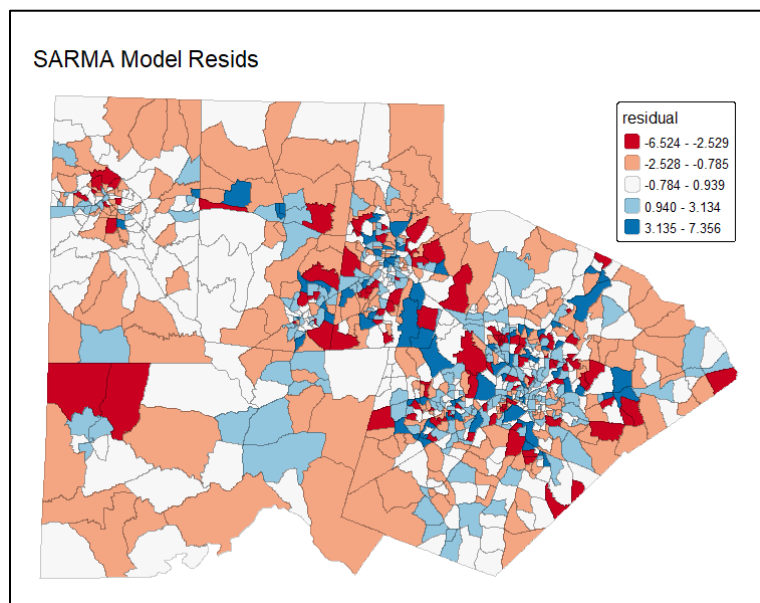
LR test value: 446.41, p-value: < 2.22e-16

Log likelihood: -1940.185 for sac model
ML residual variance (sigma squared): 5.1144, (sigma: 2.2615)
Number of observations: 831
Number of parameters estimated: 6
AIC: 3892.4, (AIC for lm: 4334.8)
```

SARMA Model.

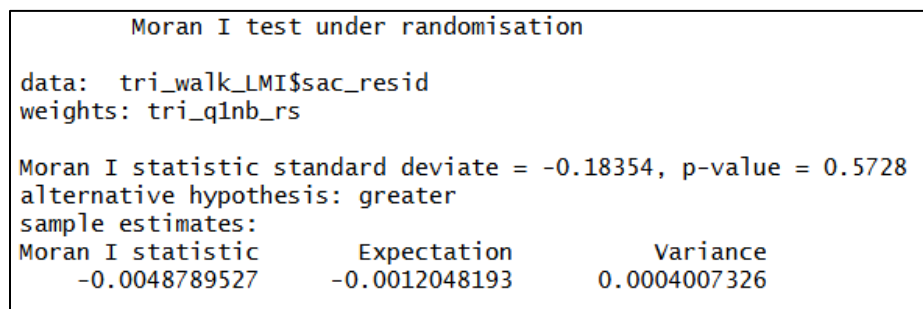
Model shows less significance (sign of less autocorrelation), but still, $p < .05$.

Verifying SARMA results



Map of SARMA residuals.

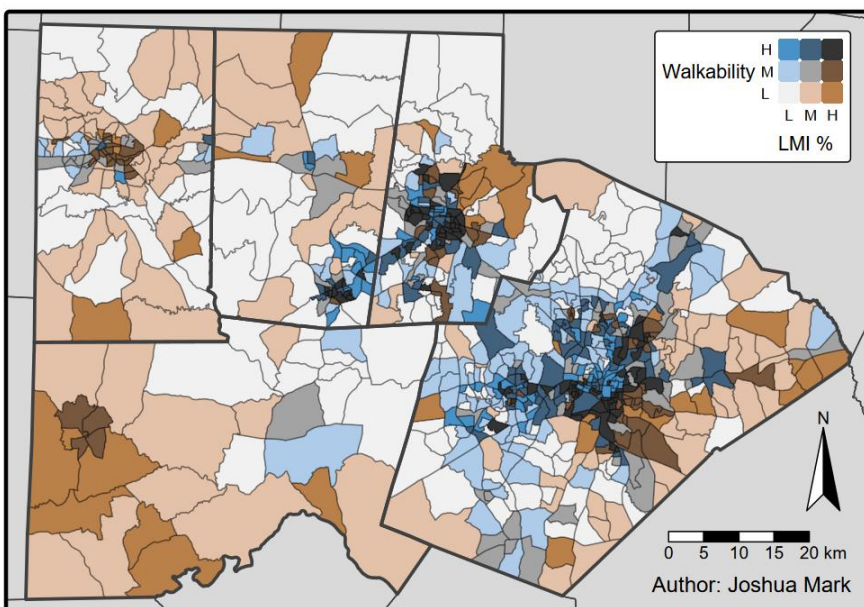
Less visible patterns
of high / low residuals.



Moran's I of residuals.

Spatial autocorrelation
is insignificant
($p > .05$).

Walkability vs. Low - Mod Income % in the NC Triangle



Final bilateral map
highlighting overlap
between walkability and
LMI %.

Categories split into low,
medium, and high based
on Fisher classifications.

Conclusions

The largest takeaway from the research is that significance was found between income and walkability, but it was in the opposite direction of the initial hypothesis. A positive association was found, meaning higher percentages of low- and moderate-income households were associated with higher walkability. This makes some sense when thought of intuitively: higher income people are more likely to live in suburban areas, often separated from city centers and sources of public transportation.

While inequity was not found in this project, more research ought to be done into this area. New research can focus on more potent quality of life indicators, like proximity to green spaces or other public services. The project can also be expanded into analyzing protected classes, like finding associations based on race.

Overall, this project dived into walkability in the NC Triangle. A spatial model was built, revealing significant predictors of walkability, both population density and the focus of the project, level of income.